

Sameer Kumar Panda

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Driven by a passion for building scalable, high-performance systems and advancing Generative AI, I bring a strong foundation in designing, developing, and deploying distributed software solutions to solve complex problems. My expertise spans architecting fault-tolerant microservices, optimizing algorithms for efficiency, and integrating Large Language Models (LLMs), Model Context Protocol (MCP), and other advanced machine learning systems into production environments. Skilled in Python, Java, C++, cloud platforms, and modern DevOps practices, I excel at delivering customer-focused, reliable, and maintainable applications—leveraging LLMs, RAG pipelines, MCP workflows, and prompt engineering to enhance intelligence, adaptability, and automation at scale.

Websites

- [LinkedIn](#)
- [GitHub](#)
- [Leetcode](#)

SKILLS

- Programming Languages:** Python, Java, C++, SQL, PySpark
- Machine Learning Algorithms:** TiDE, SARIMAX, Prophet, XGBoost, DeepAR, LSTM, Lasso Regression, Classification, Recommendation Systems, Sentiment Analysis
- Data Engineering:** ETL Pipelines, Data Preprocessing, Feature Engineering
- Cloud & DevOps:** Azure ML Studio, AWS Textract, Docker, GitHub, GitHub Actions, CI/CD, Linux, System Design
- Tools & Visualization:** Matplotlib, Seaborn, Plotly, Tableau, Power BI
- Web & API Development:** FastAPI, Flask, Django, REST APIs
- Deep Learning & NLP:** TensorFlow, PyTorch, Hugging Face, LangChain, LangGraph, LLAMA3, LoRA, DPO, Retrieval-Augmented Generation (RAG), Prompt Engineering
- Databases & Storage:** MongoDB, Redis, Vector Databases (FAISS, Weaviate)
- Testing & Optimization:** Automated Testing, Model Evaluation (Accuracy, Precision, Recall, F1-score), Performance Benchmarking, Load Testing
- Core CS Competencies:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Scalability, Fault Tolerance, Distributed Systems

Professional Experience & Impact

Machine Learning Engineer

Tiger Analytics, Hyderabad, India

October 2023 - Current

- Engineered and deployed scalable **TiDE forecasting** microservices (**SARIMAX, Prophet, XGBoost, DeepAR, LSTM**) using Python, Docker, and Azure ML—achieving up to **90% KPI accuracy gains** while ensuring reliability, observability, and global readiness.
- Architected an **agentic RAG pipeline** using **LangGraph** & **LangChain** with advanced feature engineering (**lags, seasonality, external regressors**), reducing manual forecasting effort by 95%, designed for fault tolerance and ultra-low latency.
- Built and productionized fine-tuning pipelines for **LLAMA3** (LoRA, DPO) with rigorous prompt

validation to minimize **hallucinations**, aligned with **secure and compliant deployment practices**.

- Developed RESTful microservices for AI assistants (healthcare tracking) using **FastAPI, Redis, VectorDBs**—leveraging recommendation systems and **NLP (classification, sentiment analysis)**—structured for modularity, performance, and maintainability.
- Automated **CI/CD processes** using **GitHub Actions, Docker, and Azure ML**, integrating **model validation metrics (accuracy, precision, recall, F1)**, **load testing**, and **performance dashboards** to support **robust development cycles and platform stability**.

SDE Intern

Polynomial.ai, Remote, India

February 2023 - July 2023

- Designed and deployed a distributed real-time audio-to-text pipeline using **AWS Texttract, FastAPI, and Docker**—reducing **end-to-end latency** by 90% while ensuring high throughput and fault tolerance for time-critical workloads.
- Optimized an **entity extraction microservice (TensorFlow, Python, Redis)** through algorithm refactoring and data-access optimizations, improving processing speed by 92% and enabling near-real-time document parsing at scale.
- Built a recommendation engine for automated invoice/POS data processing (**MongoDB, Django, REST APIs**), reducing manual workforce requirements by 82% and improving operational efficiency across multiple business units.
- Enhanced an **AI bot for automated property** description generation, improving **NLP model** efficiency by 82% through caching strategies, batch inference **optimization**, and **modular service design**.

Education

IIIT Bhubaneswar at Bhubaneswar, Odisha

B.Tech in Information Technology August 2023

GPA: 7.82 / 10

Projects

JOB.ai – Automated Job Aggregation & Recommendation Platform AI JOB ASSISTANT

Tech Stack: Python, FastAPI, FAISS, MongoDB, Docker, Telegram Bot API, REST APIs, Parallel Processing, Data Cleaning Pipelines

- **Designed and deployed** a distributed, fully automated job aggregation platform scraping from LinkedIn, Naukri, Instahyre, and other sources in **parallel**, reducing job data ingestion time by **99%**.
- **Built scalable ETL pipelines** for JSON normalization, data cleaning, deduplication, and schema unification—ensuring high-quality datasets for downstream processing.
- **Implemented semantic search capabilities** using FAISS vector databases for fast, context-aware job matching, enabling sub-second query responses.
- **Developed a Telegram bot interface** to deliver personalized job recommendations, integrated with MongoDB for persistent storage of user profiles, CVs, and interaction history.
- **Automated application support workflows** to generate tailored referral emails, cover letters, and cold outreach templates, improving candidate application efficiency by **~80%**.
- **Containerized and orchestrated** the entire system using Docker for reproducible deployments and high availability in production.

Accomplishments

- **2nd Place – Tiger Analytics Agentic AI Hackathon** – Built and deployed an advanced Agentic AI workflow integrating LLMs, RAG, and automation pipelines for real-world business use cases.
- **5★** – Cleared all milestones in the Tiger Analytics MLE Program.
- **1585 / 34K** – Facebook Hacker Cup (Round 2).
- **1996th & 2448th** – Google Kickstart (Round F & B).
- **144th / 8K** – CodeChef Short Challenge.